

of events. The configuration can also be customised for fine-tuning the sensitivity and ranking of behaviour.

Microsoft Machine Learning for Apache Spark (MMLSpark)

Microsoft Machine Learning for Apache Spark comprises tons of tools aimed at expanding the distributed computing framework Apache Spark in diverse directions. It adds deep learning and data science tools to Spark and even integrates Spark machine learning pipelines with the Microsoft Cognitive Toolkit (CNTK), LightGBM and OpenCV. This helps application designers to create powerful, scalable, predictive and analytical models for large images and text based data sets.

Some of its features are:

- Preprocessing of image data sets using OpenCV
- Strong integration of images from HDFS into Spark
- Image training using pretrained deep neural nets using CNTK
- Web services in Spark using HTTP on Apache Spark
- Classification and regression models training

Azure Language Understanding (LUIS)

Azure Language Understanding (LUIS) allows an application to understand what a person wants. It uses machine learning to help developers design applications that accept the input in natural language and extract the meaning from it. The application interacts with end users to pass the input into the LUIS app and translate it into relevant information to fetch the results back.

The technologies that LUIS works with are:

- *Bot framework*: This chatbot based API talks with the end user via a text input.
- *Bing Speech API*: This converts the natural language into text and then

LUIS processes it.

There are three key concepts on which LUIS is based.

- **Intents**: An intent represents the actions a user wants to perform. The intent is a purpose or goal expressed in a user's input, such as booking a flight, paying a bill, or finding a news article. You define and name intents that correspond to these actions. A travel app may define an intent named 'BookFlight'.
- **Utterances**: An utterance is text input from the user that your app needs to understand. It may be a sentence, like 'Book a ticket to Paris', or a fragment of a sentence, like 'Booking' or 'Paris flight'. Utterances aren't always well-formed, and there can be many utterance variations for a particular intent.
- **Entities**: An entity represents detailed information that is relevant in the utterance. For example, in the utterance 'Book a ticket to Paris', 'Paris' is a location. By recognising and

labelling the entities that are mentioned in the user's utterance, LUIS helps you to choose the specific action to take to answer a user's request.

LUIS has two ways to build a model — the Authoring APIs and the LUIS.ai Web app. Both methods give developers and their collaborators control of the LUIS model definition. One can use either LUIS.ai or the Authoring APIs or a combination of both to build a model. This management includes models, versions, collaborators, external APIs, testing, and training. Once the model is built and published, the utterance is passed to LUIS and the JSON object results are received with the endpoint APIs.

Microsoft Azure has introduced a range of AI and ML services in recent years to help developers and businesses tackle the large amounts of data being generated. Do take some time out to learn more about them (check the *References* below) to keep up with these fast changing times. 

References

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By: Dr Anand Nayyar and Dr Magesh Kasthuri

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.